CHANDIGARH UNIVERSITY

C PROGRAMMING PROJECT REPORT

QUIZ APPLICATION
USING C PROGRAMMING

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Subject	Computer Programming (C)
Academic Year	2026–2027

Submitted in partial fulfilment of the C Programming project work.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my teacher, Preeti, for providing me with the opportunity to develop this C Programming project titled “Quiz Application Using C.” This project helped me understand how the concepts learned in class can be applied to create a working program.

I am thankful for the guidance and support that helped me organize the project and understand the logic behind displaying questions, checking answers, and calculating the final score.

I would also like to thank Chandigarh University for providing a learning environment that encourages practical knowledge and programming skills. This project has been a useful learning experience and has improved my confidence in writing and testing C programs.

1. INTRODUCTION

A Quiz Application is a computer-based program developed using the C programming language. It allows users to participate in a quiz by answering multiple-choice questions on different topics. The application displays questions along with four options, accepts the user's answer, and checks whether the selected answer is correct or incorrect.

The program maintains the user's score throughout the quiz and displays the final result after all questions have been answered. It provides a simple and interactive way to test knowledge and improve learning.

This project demonstrates important C programming concepts, including variables, loops, conditional statements, and input/output operations. The questions are handled in sequence, and the score is updated whenever the user gives a correct answer. The project can be extended in the future by storing questions in arrays, adding categories, or allowing users to play again.

2. AIM OF THE PROJECT

The main aim of this project is to design and develop a simple, interactive Quiz Application using the C programming language. The application allows users to answer multiple-choice questions, checks their responses, calculates their score, and displays the final result.

The project also aims to improve programming knowledge and develop logical thinking through the practical implementation of C programming concepts.

Expected Result

At the end of the quiz, the application should display the total score out of five and show a short performance message based on the score obtained.

3. OBJECTIVES OF THE PROJECT

The main objectives of the Quiz Application are:

• To develop an interactive quiz program using C programming.

• To display multiple-choice questions with four answer options.

• To accept answers from the user through keyboard input.

• To check the correctness of each answer using conditional statements.

• To calculate and maintain the user's score.

• To display the final score after completing the quiz.

• To understand the practical use of variables, input/output functions, and decision-making statements.

• To improve logical thinking, problem-solving, and program-testing skills.

4. HARDWARE AND SOFTWARE REQUIREMENTS

A. Hardware Requirements

• Computer or laptop: Used to write, compile, and execute the program.

• Processor: A basic processor capable of running a C compiler.

• RAM: At least 2 GB of RAM is sufficient for this basic program.

• Keyboard: Used to enter answers and interact with the application.

B. Software Requirements

• Operating System: Windows, Linux, or another operating system that supports a C compiler.

• Programming Language: C.

• Code Editor: Visual Studio Code (VS Code) for writing and editing the source code.

• C Compiler: GCC or another compatible C compiler to compile and execute the program.

These requirements are sufficient for developing and running this console-based application.

5. C PROGRAMMING CONCEPTS USED

1. Variables

Variables store information such as the user's selected answer and total score. In this project, integer variables are used for answer input and score calculation.

2. Functions and main()

The main() function is the starting point of the program. The program statements are arranged inside main() to control the quiz from beginning to end.

3. Conditional Statements

The if statement checks whether the selected answer matches the correct option. If it matches, the score increases by one. The if-else-if structure is also used to display a performance message.

4. Input and Output

The printf() function displays questions, options, and results. The scanf() function accepts the user's answer from the keyboard.

5. Operators and Score Calculation

The equality operator (==) compares the user's answer with the correct answer. The increment operator (++) increases the score when an answer is correct.

Note: This version focuses on variables, input/output, conditional statements, and score calculation. Arrays, loops, and separate user-defined functions can be added in an enhanced version.

6. ALGORITHM

The algorithm describes the logical sequence followed by the Quiz Application.

Step 1: Start the program.

Step 2: Initialize the score to zero.

Step 3: Display the quiz title and the first question with four options.

Step 4: Ask the user to enter an option from 1 to 4.

Step 5: Compare the entered answer with the correct answer.

Step 6: If the answer is correct, increase the score by one; otherwise, leave the score unchanged.

Step 7: Repeat the question-and-answer process for all five questions.

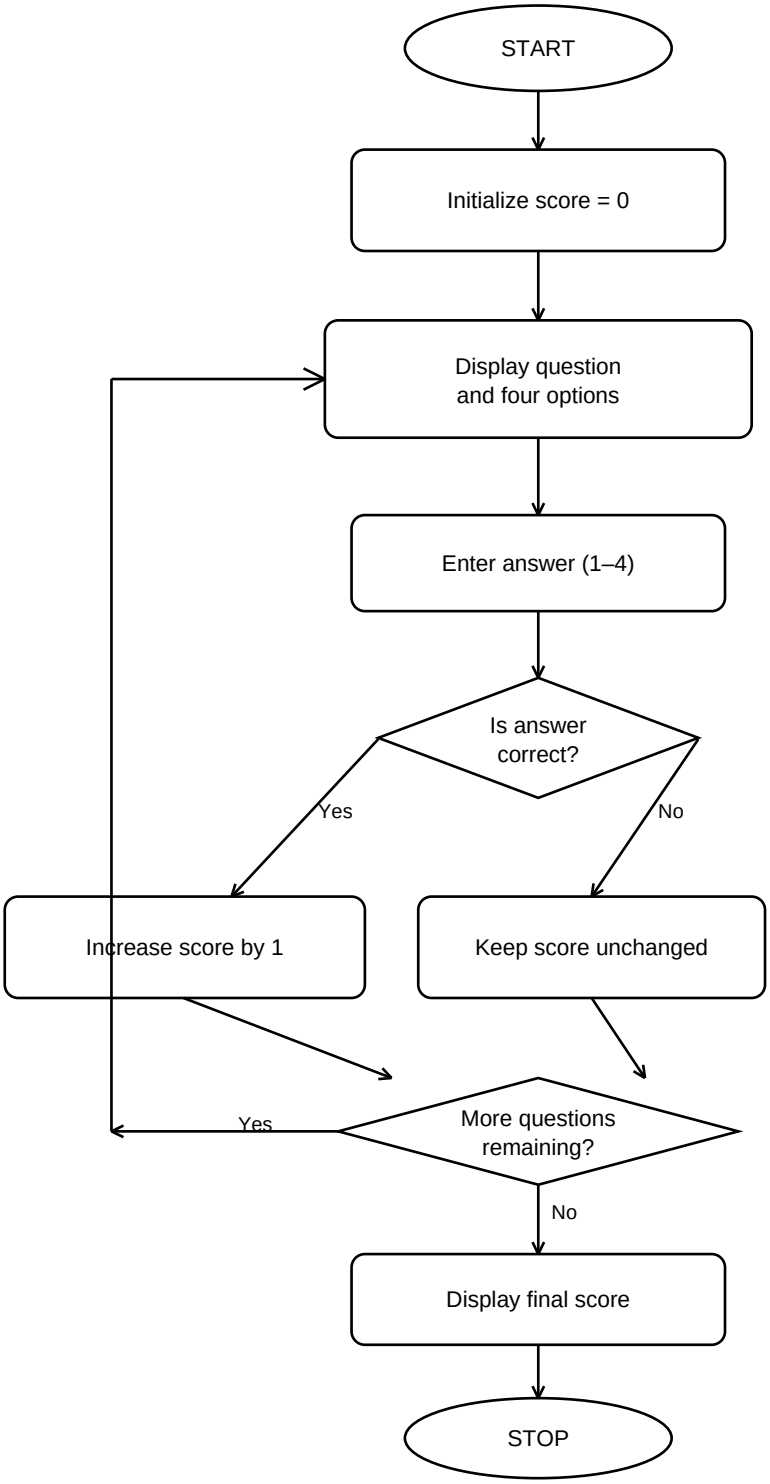
Step 8: Display the final score out of five.

Step 9: Display a performance message based on the score.

Step 10: Stop the program.

7. FLOWCHART

The flowchart below shows the sequence of operations performed by the Quiz Application, from starting the program to displaying the final score.



8. SOURCE CODE

The following C program presents five multiple-choice questions, accepts an answer for each question, checks the answer, and calculates the final score.

```
#include <stdio.h>

int main()
{
    int answer, score = 0;

    printf("================================\n");
    printf("       QUIZ APPLICATION\n");
    printf("================================\n");

    // Question 1
    printf("\n1. Which language is used for C programming?\n");
    printf("1. Python\n2. C\n3. Java\n4. HTML\n");
    printf("Enter your answer (1-4): ");
    scanf("%d", &answer);

    if (answer == 2)
        score++;

    // Question 2
    printf("\n2. Which function displays output in C?\n");
    printf("1. scanf()\n2. main()\n3. printf()\n4. return\n");
    printf("Enter your answer (1-4): ");
    scanf("%d", &answer);

    if (answer == 3)
        score++;

    // Question 3
    printf("\n3. Which symbol ends a statement in C?\n");
    printf("1. .\n2. ,\n3. :\n4. ;\n");
    printf("Enter your answer (1-4): ");
    scanf("%d", &answer);

    if (answer == 4)
        score++;

    // Question 4
    printf("\n4. Which loop executes while a condition is true?\n");
    printf("1. while\n2. switch\n3. if\n4. break\n");
    printf("Enter your answer (1-4): ");
    scanf("%d", &answer);

    if (answer == 1)
        score++;

    // Question 5
    printf("\n5. Which header file is required for printf()?\n");
    printf("1. math.h\n2. stdio.h\n3. string.h\n4. conio.h\n");
    printf("Enter your answer (1-4): ");
    scanf("%d", &answer);

    if (answer == 2)
        score++;

    printf("\n================================\n");
    printf("          QUIZ RESULT\n");
    printf("================================\n");
    printf("Your Score: %d out of 5\n", score);
```

```
    if (score >= 4)
        printf("Excellent Performance!\n");
    else if (score >= 2)
        printf("Good Try! Keep Practicing.\n");
    else
        printf("Keep Learning and Try Again!\n");

    printf("Thank you for participating!\n");

    return 0;
}
```

9. SAMPLE OUTPUT

The following is an example of how the program may appear when it is executed. The sample demonstrates the quiz title, question format, answer entry, and final result.

```
================================
       QUIZ APPLICATION
================================

1. Which language is used for C programming?
1. Python
2. C
3. Java
4. HTML
Enter your answer (1-4): 2

2. Which function displays output in C?
1. scanf()
2. main()
3. printf()
4. return
Enter your answer (1-4): 3

... Remaining questions appear in the same format ...

================================
          QUIZ RESULT
================================
Your Score: 5 out of 5
Excellent Performance!
Thank you for participating!
```

Note: This is illustrative sample output. The score and performance message will change depending on the answers entered by the user.

10. CONCLUSION

The Quiz Application was developed using the C programming language. It allows users to answer multiple-choice questions and calculates their scores automatically. The project demonstrates the use of variables, input/output functions, conditional statements, comparison operators, and basic decision-making.

Through this project, I gained practical knowledge of program development and learned how to apply programming concepts to solve a simple problem. Testing the program also helped me understand how user input affects the result.

The application can be improved in the future by using arrays and functions to organize questions, adding more questions or categories, validating incorrect input, and providing an option to restart the quiz.

11. LEARNING OUTCOMES

After completing this project, I learned:

• How to write, compile, and execute a C program.

• How to declare and initialize variables.

• How to accept user input using scanf().

• How to display questions and results using printf().

• How to use if and if-else-if statements for decision-making.

• How to compare answers and update a score.

• How to test a program and verify its output.

• How programming logic can be used to create a simple interactive application.

This project strengthened my understanding of basic C programming and provided practical experience in developing a console-based application.